▯ CIRCULATORY SYSTEM — FULL MIND MAP

Biology Ch. 7.v · Human Anatomy & Physiology · Hinglish · SSC / BPSC / BSSC / Railway

📄 Topics in this sub-chapter:

1. Heartbeat, Cardiac Cycle & Pulse

- Healthy person ka normal heart rate source ke according about 72–75 beats per minute hai. Cardiac cycle ek complete heartbeat ko represent karta hai; sleep ke dauran slow rates around 40–50 BPM common bataye gaye hain. *[Q1]*
- Har heartbeat mein contraction aur relaxation hota hai. Source ke mutabik heart relaxation aur contraction ke beech rest karta hai; wrist pulse bhi heart ke same rate par beat karta hai. *[Q2, Q8]*

2. Heart Attack — Typical Signs

- Heart attack ko myocardial infarction kaha gaya hai. Typical signs mein severe chest pain/discomfort, nausea aur severe perspiration/sweating shamil hain; pain/discomfort arms, back, neck, jaw ya stomach mein bhi ho sakta hai. *[Q3, Q4]*
- Source Q3 mein "pain in legs" ko heart-attack symptom nahi maana gaya, aur Q4 mein headache ko typical/direct sign nahi maana gaya. *[Q3, Q4]*

3. Heart Chambers, Electrical Impulse & Pacemaker

- Human heart mein four chambers hote hain — right/left atria aur right/left ventricles. Atria incoming blood receive karte hain aur ventricles blood pump out karte hain. *[Q5, Q6]*
- Heartbeat ka electrical impulse source "sinus node" se originate batata hai. Sino-atrial (SA) node right atrium mein hota hai aur heart ka natural pacemaker maana gaya hai. *[Q7, Q14]*
- Artificial pacemaker abnormal heart rhythm ko control/regulate karne ke liye electrical pulses use karta hai, taaki heart normal rate par beat kare. *[Q13]*

4. Heart Transplant & Artificial Heart

- Source ke mutabik world ka first human heart transplant Dr. Christian Barnard/Barnard ne 3 December 1967 ko South Africa ke Groote Schuur Hospital mein Louis Washkansky par perform kiya. *[Q9, Q10, Q11]*
- Jarvik-7 ko first successful permanent artificial heart ke roop mein describe kiya gaya hai; Robert Jarvik se associated hai aur 1982 mein patient Barney Clark mein first implantation ka reference diya gaya hai. *[Q12]*

5. Pulmonary Vein & Coronary Arteries

- Pulmonary veins body ki only veins batayi gayi hain jo oxygenated blood lungs se left atrium tak carry karti hain. Source ye bhi note karta hai ki pulmonary arteries oxygen-poor blood carry karti hain. *[Q15]*
- Heart muscle (myocardium) ko oxygen-rich blood supply karne wali vessels coronary arteries hoti hain; deoxygenated blood remove karne wali vessels cardiac veins batayi gayi hain. *[Q78]*

6. Blood — Volume, Nature, Plasma & Functions

- 70 kg normal adult mein total blood volume source ke according roughly 5–6 litres hai. *[Q16, Q17]*
- Blood ek special connective tissue hai jiska fluid matrix plasma hota hai. Plasma blood ka fluid part hai; blood vessels ke andar blood oxygen, carbon dioxide aur nutrients transport karta hai. *[Q18, Q20, Q31]*
- Q19 ka source answer erythrocytes (RBCs) select karta hai; explanation RBCs ko haemoglobin ke through oxygen tissues tak aur carbon dioxide tissues se lungs tak carry karne se link karti hai. *[Q19]*

⚠ EXAM TRAP — Q17 Disputed Blood-Volume Question

▸ Q17 mein source answer label "Disputed Question (Ans. *)" hai. Options mein 6000 ml aur 5000 ml dono diye gaye hain, jabki explanation normal 70 kg adult ka total blood 5–6 litres batati hai; isliye source kisi ek option ko final correct mark nahi karta. *[Q17]*

7. Haemoglobin, RBCs, Myoglobin & Red Colour

- Haemoglobin RBCs mein present red, iron-containing protein hai jo lungs se oxygen tissues tak carry karta hai aur source explanation carbon dioxide ko tissues se lungs ki taraf le jane se bhi link karti hai. "Slightly acidic" statement ko Q22 mein incorrect maana gaya hai. *[Q21, Q22, Q25]*
- RBC-related correct properties: iron present hota hai, blood ko red colour milta hai aur oxygen transport hota hai; immunity dena RBC/haemoglobin ka function nahi maana gaya. *[Q23, Q24]*
- Myoglobin skeletal muscle ka oxygen-binding heme protein hai aur iron (Fe) contain karta hai. *[Q26, Q27]*
- Earthworm mein haemoglobin plasma mein dissolved hota hai. Human blood/RBC ka red colour haemoglobin se linked hai. *[Q28, Q29, Q30]*

8. Blood Pressure — Normal Value, Instrument & Age

- Normal average blood pressure 120/80 mm Hg diya gaya hai — around 120 mm Hg systolic aur 80 mm Hg diastolic. Given examples mein 120/80 ko normal select kiya gaya hai. *[Q32, Q33, Q38]*
- Blood pressure measure karne ka instrument sphygmomanometer hai; "Hg" mercury ka chemical symbol hai. *[Q34, Q35, Q36]*
- Source ke according age badhne par blood pressure generally increase hota hai; older age ko high blood pressure ke contributing factors mein include kiya gaya hai. *[Q37]*

9. Source-Specific Blood Pressure vs Atmospheric Pressure

- Q39 source blood pressure ko atmospheric pressure se greater select karta hai. Explanation ke mutabik clinical blood-pressure reading atmospheric pressure ke relative hoti hai; 120 mm Hg systolic ko atmospheric pressure se 120 mm Hg above aur vacuum reference par roughly 760 + 120 = 880 mm Hg ke framing mein explain kiya gaya hai. *[Q39]*

10. Blood Groups — Discovery, Antigens & Rh Factor

- Karl Landsteiner ko main blood groups distinguish/discover karne se credit diya gaya hai. Source Rh factor ko Rhesus monkey se naam mila batata hai aur Landsteiner + Alexander S. Wiener ko Rh factor identification se associate karta hai. *[Q40, Q41]*

- Blood group RBC surface par specific antigens ki presence/absence se determine hota hai. ABO system mein A, B, AB aur O types A/B antigens ke basis par bante hain; Q42 ke listed options mein correct determinant present na hone ke karan “None of the above” answer hai. *[Q42]*
- Blood group O ke RBCs par neither A nor B agglutinogen/antigen hota hai. *[Q43]*

11. Universal Donor / Recipient & Transfusion Rules

- Source ABO framing mein blood group O ko universal donor maanta hai because RBC surface par A/B antigens nahi hote. Jab unknown blood group ko emergency transfusion dena ho, Q52–Q53 specifically O negative / O, Rh− ko safe option select karte hain. *[Q44, Q45, Q46, Q52, Q53]*
- Blood group AB ko universal recipient/acceptor maana gaya hai because plasma mein anti-A/anti-B antibodies absent hote hain; AB person source ke according any ABO group se blood receive kar sakta hai. Q47 mein Assertion true aur Reason false hai, kyunki AB RBCs par A aur B dono antigens present hote hain. *[Q47, Q48, Q49, Q50]*
- Blood group A individual source ke mutabik A ya O group se blood receive kar sakta hai; accident scenario mein wife O, son A aur daughter O donate kar sakte hain, brother AB nahi. *[Q51]*

12. Blood-Group Inheritance & Rh Incompatibility

- AB+ × O− parents ke example mein source biological twin sons ko A+ aur B+ assign karta hai; therefore three sons mein O+ child adopted maana gaya hai. *[Q54]*
- Father group A aur mother group O ke case mein source explanation offspring ko A ya O possible batati hai aur ABO inheritance ko Mendelian genes se link karti hai. *[Q55]*
- Transfusion mein ABO ke saath Rh factor compatibility bhi important hai. Rh antigen present ho to Rh+ aur absent ho to Rh−; Rh− person ko Rh+ blood exposure par antibodies form ho sakti hain. *[Q56]*
- Maternal–foetal Rh incompatibility ka classic source case Rh− mother aur Rh+ foetus hai; explanation erythroblastosis fetalis/hemolytic anaemia se link karti hai. *[Q57]*

> **⚠ EXAM TRAP — Q55 Answer Key vs Explanation**
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> ▸ Q55 ki answer key option (C) = O select karti hai, lekin same source explanation clearly kehti hai ki father A + mother O hone par son ka blood group “O or A” ho sakta hai. Is internal source mismatch ko as-is preserve kiya gaya hai. *[Q55]*

13. Blood pH

- Normal human blood ka pH about 7.4 hai; source range 7.35–7.45 deta hai, yani blood slightly basic/alkaline hota hai. *[Q58, Q59, Q60]*

14. Antigen, WBCs, Lymphocytes & Immunity

- Antigen generally foreign molecule/substance hai jo immune system trigger karke antibody formation induce karta hai. *[Q61, Q62, Q63]*
- WBCs infection se protection aur disease resistance mein major role play karte hain. *[Q64, Q65]*
- Lymphocytes antibodies produce karte hain aur immunity se closely related hain; source lymphocytes mein B cells, T cells aur natural killer cells ka reference deta hai. *[Q66, Q67, Q68]*
- B and T lymphocytes pathogens se hone wali diseases ke against adaptive immune response ka critical part hain. Vaccination ke through antigen introduce hone par source Q70 antibodies banane ke liye B cells ko primary cells select karta hai. *[Q69, Q70]*

15. Blood Viscosity, WBC Size & Leukemia

- Source blood viscosity ko mainly hematocrit aur plasma viscosity se relate karta hai; Q71 ka answer “proteins in blood” hai because plasma proteins plasma viscosity ko affect karte hain. *[Q71]*

- WBC diameter source mein about 0.007 mm diya gaya hai; normal WBC count roughly 4,000–11,000 per microlitre aur total blood volume ka approx 1% bataya gaya hai. *[Q72]*
- Q73 source excess abnormal white corpuscles ke pathological condition ko leukemia select karta hai; explanation leukemia ko bone marrow se start hone wale cancers ke group se link karti hai. *[Q73]*

16. Blood-Cell Formation, Spleen & Lymphocytes

- Adult mein RBCs red bone marrow of large bones mein continuously form hote hain (erythropoiesis); source embryo mein liver ko main RBC-production site batata hai aur kidney-made erythropoietin (EPO) ko stimulation se link karta hai. *[Q74]*
- Spleen old RBCs remove/destroy karta hai aur immune/blood-filter role rakhta hai. Source Q75 spleen ko WBC formation + RBC destruction ke option mein select karta hai, while explanation adult bone marrow ko all RBCs, 60–70% WBCs and all platelets produce karne wala bhi batati hai. *[Q75]*
- Lymphocyte production source explanation multiple lymphatic tissues — spleen, thymus, lymph nodes — aur bone marrow context se jodti hai. Isi wajah se Q76 spleen select karta hai, jabki Q77 “more than one of the above” answer deta hai. *[Q76, Q77]*

17. Blood Clotting, Plasma Water & Platelets

- Thrombin blood clotting/coagulation se associated enzyme hai jo soluble fibrinogen ko insoluble fibrin strands mein convert karta hai. *[Q79, Q80]*
- Plasma blood volume ka about 55% hai aur source ke according roughly 91–92% water hota hai; Q81 ka closest listed answer 90% hai. Plasma mein albumin, fibrinogen aur globulins/antibodies bhi mentioned hain. *[Q81]*
- Platelets clotting mein injury site par aggregate/stick karke coagulation platform banate hain. Source normal platelet count about 150,000–350,000/μL, RBC ~5,000,000/μL aur WBC ~4,000–11,000/μL deta hai; isliye blood mein WBC se platelets zyada hote hain. *[Q82]*
- Alum ko cuts se bleeding stop karne ke liye source aluminium ions ki high coagulating power aur blood ke colloidal nature se explain karta hai; Assertion aur Reason dono true aur Reason correct explanation maana gaya hai. *[Q84]*

18. Nitric Oxide — Vasodilation & Blood Flow

- Nitric oxide (NO) human body mein synthesize hota hai aur blood-vessel smooth muscle relax karke vasodilation karta hai, jisse blood flow increase aur blood pressure lower ho sakta hai. Source endothelium ko primary synthesis site batata hai. *[Q83]*

19. Artificial “Plastic Blood”

- Source University of Sheffield, Britain ke scientists ko artificial “plastic blood” development se associate karta hai. Is substitute ko emergency/military use ke liye light, easy to store, refrigeration-free aur water mein dissolve karke use karne ke concept ke roop mein describe kiya gaya hai. *[Q85]*